

Jiten Dhandha

PhD student - University of Cambridge

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arXiv: [dhandha_j-1](https://arxiv.org/archive/journal/journal/00000002/00000002-1481-0907)
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Research interests

My research focuses on the study of the early Universe, spanning the **Dark Ages, Cosmic Dawn, and the Epoch of Reionization** (redshift $z \sim 6 - 50$). My expertise lies in the theoretical modelling of the **cosmic 21-cm signal** of neutral hydrogen, and of **stellar and galactic astrophysics**; this includes numerical simulations of present-day ($z \sim 0$) Population I star-formation in molecular clouds, as well as analytic and semi-numerical simulations of high-redshift ($z \gtrsim 6$) Population II and Population III stars. I have extensively used **machine learning and Bayesian inference techniques** to accelerate simulations and perform parameter inference, combining data from multi-wavelength facilities (e.g. from JWST and 21-cm experiments).

Education

2022 - <i>present</i>	PhD in Astronomy , Institute of Astronomy, University of Cambridge. Funded by Boustany Astronomy Scholarship & Isaac Newton Studentship at Pembroke College. Supervised by Prof. Anastasia Fialkov and Dr. Eloy de Lera Acedo.
2018 - 2022	MPhys. Physics with Astrophysics First Class, University of Manchester. Project involved simulating turbulent molecular clouds in ISM and studying filament and star formation. Performed with Zoe Faes and supervised by Dr. Rowan Smith.
2016 - 2018	All India Senior School Certificate Examination , DPS - Modern Indian School, Doha, Qatar. Average of 95.2% in AISSCE examination.

Employment

Jul. 2024 - Aug. 2024	Summer volunteer internship , Boustany Foundation, Monaco. Partnering with Open Cultural Center, a humanitarian NGO focused on providing teaching, advice and extracurricular activities to asylum-seekers and refugees in the Nea Kavala camp in Greece.
Jun. 2021 - Aug. 2021	Summer research project , University of Manchester. Modelling the cosmological 21-cm signal in Recfast++ and CosmoTherm to study their synergy with CMB spectral distortions. Supervised by Prof. Jens Chluba.
Jun. 2020 - Sep. 2020	Summer research project , University of Manchester. Testing and debugging LOFAR-VLBI calibration/imaging pipeline for gravitational lenses. Supervised by Dr. Neal Jackson.
Jul. 2019 - Sep. 2019	Summer intern programme , British Petroleum / University of Manchester. Simulating mitigation techniques for sulphate reducing bacteria responsible for fouling crude oil. Supervised by Dr. Thomas Waigh.

Grants and awards

April 2024	DiRAC Resource Allocation Committee 16th Call , awarded 4.15M CPUh (worth £41,500) on DiRAC's COSMA-8 supercomputer.
July 2022	Tessella Prize for Software (£125), for outstanding work implementing software in Mphys project.
March 2022	Boustany Scholarship for Astronomy ($\sim$ £100,530), for PhD at University of Cambridge.
April 2019	BP Achievement Award (£1000), for best essay on petrophysical logging tools.
December 2018	Physics Success Scholarship (£2000), for academic excellence in physics and maths.

Services

Conference, Workshop and Meeting organization

March 2026 - <i>present</i>	21-cm Meeting Series, co-organiser of bi-weekly meetings at KICC, University of Cambridge, focused on the cosmological 21-cm signal and related topics.
May 2026	Galaxies and the 21-cm Frontier: Unveiling the Epoch of Reionization (Kavli Science Focus), member of Organising Committee and session chair, KICC, University of Cambridge.
July 2025	Radio cosmology and science with the 21-cm signal (one day workshop), member of Organising Committee and session chair, KICC, University of Cambridge.
February 2024	Science with the 21-cm line (Kavli Science Focus), member of Organising Committee and session chair, KICC, University of Cambridge.

Other committees

Feb. 2025 - Mar. 2025	International Womens Day Committee member, Institute of Astronomy, UoC.
May 2023 - <i>present</i>	EDI Inclusion and Fairness subgroup member, Institute of Astronomy, UoC.
Jul. 2023 - Jul. 2024	Graduate Parlour , Ethnic Minorities officer, Pembroke College, UoC.
Oct. 2022 - <i>present</i>	Postgraduate Forum representative, Institute of Astronomy, UoC.
Oct. 2022 - Apr. 2023	Pembroke Papers committee memeber, Pembroke College, UoC.

Teaching responsibilities

Oct. 2025 - <i>present</i>	Co-Supervision: Saughn Sekhon (Part III student) with Prof. Anastasia Fialkov and Dr. Peter Sims. Work on machine learning accelerated Bayesian inference of early Universe star-formation.
Oct. 2025 - <i>present</i>	Co-Supervision: Seowon Jung (Part III student) with Prof. Anastasia Fialkov and Prof. Alexander Belyaev. Work on probing the nature of dark matter with 21-cm cosmology.
Oct. 2024 - Jul. 2025	Co-Supervision: Jacques Valkenberg (MPhil. student) with Prof. Anastasia Fialkov and Prof. Sandro Tacchella. Work on studying the impact of metal enrichment and Population II stellar IMF on the 21-cm signal of neutral hydrogen.
Oct. 2024 - Jul. 2025	Co-Supervision: Kyle Wong (Part III student) with Prof. Anastasia Fialkov. Work on studying the impact of varying cosmology and matter power spectrum on the 21-cm signal of neutral hydrogen.
Oct. 2023 - Jul. 2024	Co-Supervision: Jamie Incley (Part III student) with Prof. Anastasia Fialkov. Work on comparison of Epoch of Reionization in simulation codes 21cmSPACE and C2-Ray .
Feb. 2023 - Mar. 2023	Demonstration of Part IA Scientific Computing for 22 hours, University of Cambridge.

Talks

Invited talks

April 2026	<i>Constraining Galaxies in the Early Universe using 21-cm and Multi-Wavelength Synergies</i> (virtual), State of the Universe (SOTU) seminar, DTP, TIFR-Mumbai.
November 2025	<i>Narrowing the discovery space of the cosmological 21-cm signal using multi-wavelength constraints</i> , Weekly seminar, Institut d'Astrophysique Spatiale, Université Paris-Saclay.
September 2025	<i>An overview of 21-cm cosmology and how we can constrain the discovery space of the 21-cm signal</i> (virtual), 8th Global 21-cm Workshop, California Institute of Technology.
October 2022	<i>Can accreting primordial black holes explain the excess radio background?</i> (virtual), Weekly webinar, PDAT Laboratory, K. N. Toosi University of Technology.

Conference and Workshop talks

March 2026	<i>Narrowing the discovery space of the cosmological 21-cm signal using multi-wavelength constraints</i> , Cosmology in the Alps 2026, Les Diablerets, Switzerland.
June 2025	<i>Exploiting synergies between JWST and cosmic 21-cm observations to uncover star formation in the early Universe</i> , SKAO General Science Meeting 2025, Görlitz, Germany.
October 2024	<i>Constraining star-formation efficiency in the early Universe using JWST and the cosmic 21-cm signal</i> , Introduction to KICC, Kavli Institute for Cosmology, University of Cambridge.

September 2024	<i>Constraining star-formation efficiency in the early Universe using JWST and the cosmic 21-cm signal</i> , 7th Global 21-cm Workshop, Raman Research Institute.
May 2024	<i>Synergies between 21-cm experiments and JWST observations</i> , Reionization in Relic Radiation (R3), Institut d'Astrophysique Spatiale, Université Paris-Saclay.
February 2024	<i>Synergies between 21-cm experiments and JWST observations</i> , One day workshop: Science with the 21-cm line, Kavli Institute for Cosmology, University of Cambridge.
September 2023	<i>FIlamentary Structure Analysis (fiesta)</i> , AREPO-ISM workshop, University of Manchester.

Meeting talks

April 2026	<i>Narrowing the discovery space of the cosmological 21-cm signal using multi-wavelength constraints</i> , NenuFAR Cosmic Dawn plenary, Kavli Institute for Cosmology, University of Cambridge.
October 2025	<i>An overview of 21-cm cosmology and how we can constrain the discovery space of the 21-cm signal</i> , mini-workshop on 21-cm signal (headed by Prof. Naoki Yoshida), Department of Physics, University of Tokyo.
September 2025	<i>What do we know about the 21-cm signal so far?</i> , Gravitational Wave Astrophysics Group group meeting (headed by Prof. Michela Mapelli), Centre for Astronomy of Heidelberg University.
September 2023	<i>Bringing 21-cm simulations to the JWST era</i> , 2nd REACH Annual Meeting, University of Malta.

Outreach talks

October 2022	<i>Like beads on a string... Where do massive stars in our Universe come from? A brief look into studying our cosmos</i> , Pembroke Papers, Pembroke College, University of Cambridge.
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Workshops attended

September 2025	IWR School on Machine Learning for Fundamental Physics , Faculty of Mathematics and Computer Science, Heidelberg University.
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Software development

eor_limits	Author and maintainer of the new beta-release of the web-app: eorlimits.streamlit.app , which allows users to compare upper limits on the 21-cm power spectrum from different experiments.
21cmSimulators	Main author and maintainer: Community-led and comprehensive public repository of the widely used 21-cm signal simulation codes.
21cmExperiments	Main author and maintainer: Community-led and comprehensive public repository of past & ongoing 21-cm experiments.
CFit	Main author and maintainer: Smart curve fitting tool using method of least squares in Python.
fiesta	Main author and maintainer: Toolkit for analyzing filament networks and density field meshes.

In the media

August 2021	Most detailed-ever images of galaxies revealed using LOFAR . Press release for LOFAR observations from ASTRON.
August 2021	Astronomers develop novel way to 'see' first stars through fog of early Universe . Press release for LOFAR observations from BBC.

Extracurricular activities

Sep. 2021 - Jul. 2022	Student Representative representing astronomy/astrophysics, UoM.
Jul. 2020 - Jul. 2022	Touch Rugby Society , Inclusion officer and COVID-19 safety officer, UoM.
Sep. 2019 - Jun. 2020	Peer-Assisted Study Session leader, Peer Support Scheme, UoM.
Nov. 2016 - <i>present</i>	English Wikipedia , volunteer editor.

Publications

I have **3 first author** publications and **14 contributing author** publications.

First Author

December 2025	J. Dhandha , A. Fialkov, T. Gessey-Jones, H. T. J. Bevens, S. Tacchella, S. Pochinda, E. de Lera Acedo, S. Singh, R. Barkana, Narrowing the discovery space of the cosmological 21-cm signal using multi-wavelength constraints , Monthly Notices of the Royal Astronomical Society, 544, 1608
September 2025	J. Dhandha , A. Fialkov, T. Gessey-Jones, H. T. J. Bevens, S. Tacchella, S. Pochinda, E. de Lera Acedo, S. Singh, R. Barkana, Exploiting synergies between JWST and cosmic 21-cm observations to uncover star formation in the early Universe , Monthly Notices of the Royal Astronomical Society, 542, 2292
April 2024	J. Dhandha , Z. Faes, R. J. Smith, Decaying turbulence in molecular clouds: how does it affect filament networks and star formation? , Monthly Notices of the Royal Astronomical Society, 529, 4699

Contributing Author

June 2026	S. A. K. Leeney et al., Scalable Bayesian data curation for next-generation radio experiments , arXiv e-prints, arXiv:2606.16525
April 2026	J. Wasserman, E. Zackrisson, J. Dhandha , A. Fialkov, L. Noble, S. Majumdar, Ultraviolet photon production rates of the first stars: Impact on the He II λ 1640 Å emission line from primordial star clusters and the 21-cm signal from cosmic dawn , Monthly Notices of the Royal Astronomical Society, 547, stag386
April 2026	S. Dasgupta, B. Liu, A. Fialkov, F. Deng, J. Dhandha , R. Barkana, Impact of Stochastic Pop III X-ray Binaries on the Cosmological 21-cm Signal , arXiv e-prints, arXiv:2604.11542
March 2026	A. K. Dash et al., Optimisation of calibration sources for global 21-cm experiments: the REACH case , arXiv e-prints, arXiv:2604.00105
March 2026	G. V. C. Allen et al., Circuit Modeling for In Situ 21 cm Radiometer Calibration , arXiv e-prints, arXiv:2603.15072
February 2026	Z. Abdurashidova et al., First Results from HERA Phase II , The Astrophysical Journal, 998, 33
October 2025	S. Munshi, F. G. Mertens, J. K. Chege, L. V. E. Koopmans, A. R. Offringa, B. Semelin, R. Barkana, J. Dhandha , A. Fialkov, R. Mériot, S. Sikder, A. Bracco, S. A. Brackenhoff, E. Ceccotti, R. Ghara, S. Ghosh, I. Hothi, M. Mevius, P. Ocvirk, A. K. Shaw, S. Yatawatta, P. Zarka, Improved upper limits on the 21-cm signal power spectrum at $z = 17.0$ and $z = 20.3$ from an optimal field observed with NenuFAR , Monthly Notices of the Royal Astronomical Society, 542, 2785
August 2025	B. Liu, D. Kessler, T. Gessey-Jones, J. Dhandha , A. Fialkov, Y. Sibony, G. Meynet, V. Bromm, R. Barkana, Effects of chemically homogeneous evolution of the first stars on the 21-cm signal and reionization , Monthly Notices of the Royal Astronomical Society, 541, 3113
March 2025	O. S. D. O'Hara, Q. Gueuning, E. de Lera Acedo, F. Dulwich, J. Cunnery, D. Anstey, A. Brown, A. Fialkov, J. Dhandha , A. Faulkner, Y. Liu, Uncovering the effects of array mutual coupling in 21-cm experiments with the SKA-Low radio telescope , Monthly Notices of the Royal Astronomical Society, 538, 31
February 2025	S. Pochinda, J. Dhandha , A. Fialkov, E. de Lera Acedo, Cosmological super-resolution of the 21-cm signal , Accepted at NeurIPS 2024 (Machine Learning and the Physical Sciences Workshop), arXiv:2502.00852
September 2024	O. S. D. O'Hara, F. Dulwich, E. de Lera Acedo, J. Dhandha , T. Gessey-Jones, D. Anstey, A. Fialkov, Understanding spectral artefacts in SKA-Low 21-cm cosmology experiments: the impact of cable reflections , Monthly Notices of the Royal Astronomical Society, 533, 2876
May 2024	A. Fialkov, T. Gessey-Jones, J. Dhandha , Cosmic mysteries and the hydrogen 21-cm line: bridging the gap with lunar observations , Philosophical Transactions of the Royal Society of London Series A, 382, 20230068
December 2022	S. K. Acharya, J. Dhandha , J. Chluba, Can accreting primordial black holes explain the excess radio background? , Monthly Notices of the Royal Astronomical Society, 517, 2454
February 2022	S. Badole, D. Venkattu, N. Jackson, S. Wallace, J. Dhandha , P. Hartley, C. Riddell-Rovira, A. Townsend, L. K. Morabito, J. P. McKean, High-resolution imaging with the International LOFAR Telescope: Observations of the gravitational lenses MG 0751+2716 and CLASS B1600+434 , Astronomy and Astrophysics, 658, A7

White Papers / Reports / Book Chapters

November 2025	E. de Lera Acedo et al. <i>21 cm Cosmology from Space: A UK Path via CosmoCube</i> , UK Space Frontiers 2035 (in prep.)
November 2025	J. Chluba et al. <i>Opening a New Window for Cosmology with CMB Spectral Distortions</i> , UK Space Frontiers 2035 (in prep.)
November 2025	A. Mesinger et al. <i>Synergies for the Epoch of Reionization and Cosmic Dawn</i> , Advancing Astrophysics: Preparing for Science with the SKAO (in prep.)